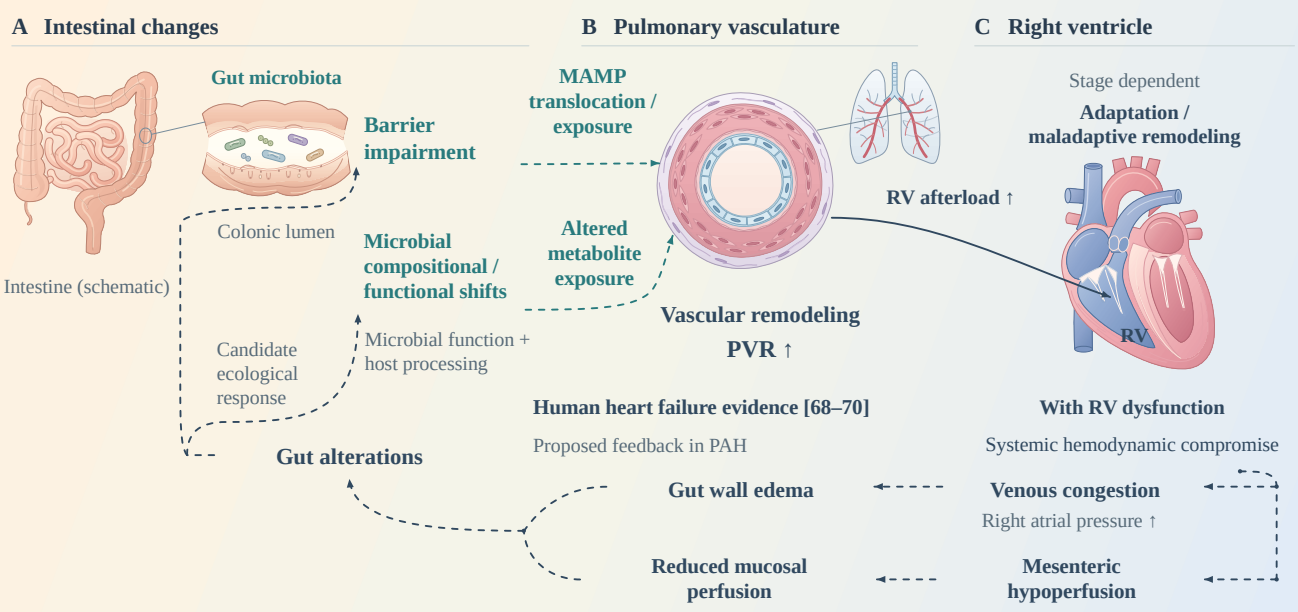

Gut–pulmonary vascular–right ventricular interactions in PAH
A Intestinal changes
B Pulmonary vasculature
C Right ventricle
Gut microbiota
Barrier impairment
Colonic lumen
Intestine (schematic)
Microbial compositional / functional shifts
Microbial function + host processing
Candidate ecological response
Gut alterations
MAMP translocation / exposure
Altered metabolite exposure
Vascular remodeling
PVR ↑
RV afterload ↑
Stage dependent
Adaptation / maladaptive remodeling
RV
Human heart failure evidence [68–70]
Proposed feedback in PAH
With RV dysfunction
Systemic hemodynamic compromise
Gut wall edema
Venous congestion
Right atrial pressure ↑
Reduced mucosal perfusion
Mesenteric hypoperfusion
L. rhamnosus GG
Experimental PH
MCT rats [71]
RV function improved
RV fibrosis reduced
Pulmonary vascular load / pulmonary arteriolar remodeling
No significant differences detected
Microbiota-related signals
Established afterload relation
Hemodynamic effects
Proposed PAH link / incomplete pathway